Task 3

Full project repo: [here]

A.

How do these specific noise frequencies affect the audio?

Adding sinusoidal noise with 500 Hz and 1000 Hz adds distinct tonal artifacts to the original base audio signal. They are not random or broadband noises but pure tones which are likely to be even more perceivable to the human auditory system—especially if they are in the frequency range over which human hearing is most sensitive.

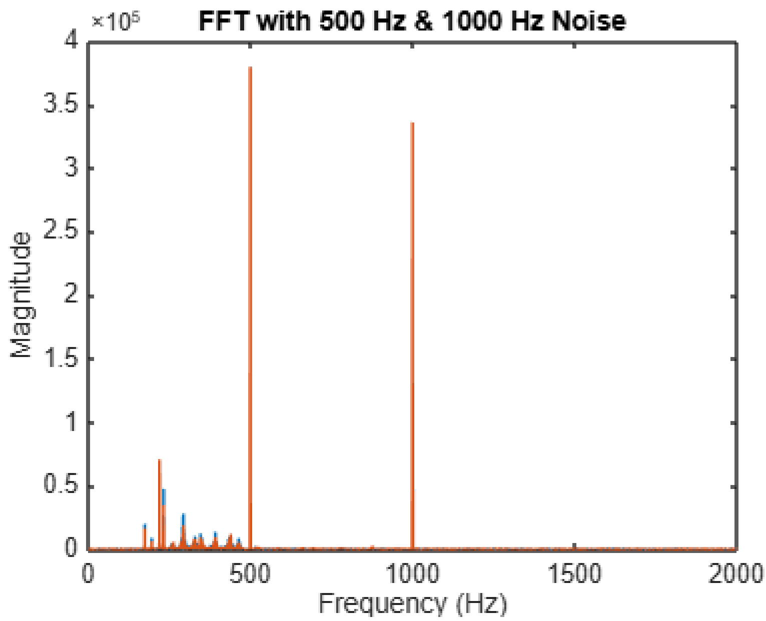
500 Hz Noise: This frequency is lower-mid and can create a buzz or hum that may interfere with instruments like cellos, pianos, or male vocalists. The noise has the form of a sustained tone and can hide or mask important musical material.

1000 Hz Noise: This is approximately the middle of sensitivity of human hearing. This noise frequency is likely to sound like a high-pitched whistle or tone, so it's particularly distracting and irritating. It gets in the way of clarity, particularly with melodic or vocal material.

B.

Can you identify the added noise frequencies in the frequency spectrum?

This is the output plot:



When you take the Fourier Transform, you're analyzing how much of each frequency is present in the signal.

By adding pure sine waves at 500 Hz and 1000 Hz, you're adding very sharp spikes in the frequency spectrum at those points. Therefore, we'll see two sharp, narrow peaks at 500 Hz and 1000 Hz.

The original audio has a complex, spread-out frequency spectrum (because music contains many tones). These clearly stand out from the rest of the audio.

C.

Which type of filter did you use, and why is it suitable for this task?

We utilized Infinite Impulse Response (IIR) Notch Filters to remove the noise we introduced.

This filter is suitable because:

1. Precise Frequency Targeting
   Notch filters are employed to remove an extremely narrow range of frequencies. Since we are aware that the noise is exactly at 500 Hz and 1000 Hz (according to the Fourier analysis), notch filters are optimal for targeting and erasing only these frequencies.

2. Minimal Distortion of Original Audio
   Unlike low-pass or high-pass filters, notch filters do not come in contact with the remaining parts of the audio spectrum. They suppress the unwanted tones without distorting or cutting vital components of the original audio.

3. Powerful and Compact
   IIR filters are computationally efficient and require fewer coefficients compared to FIR filters to produce the same performance, making them suitable for real-time or massive audio processing.

4. Adjustable Sharpness (Q factor)
   The notch filter contains a parameter called the Q factor, which specifies the narrowness or width of the notch. The higher the Q gives an exceedingly sharp, well-defined cut—perfect for cutting out narrowband sinusoidal interference like the 500 Hz and 1000 Hz tones that you added.

5. According to Frequency Analysis (FFT)
   Since the frequencies of the noise could readily be observed on the FFT display as separate spikes, a filter that directly and surgically deletes them (i.e., a notch filter) is the most suitable and effective solution.

The IIR notch filter is thus the best choice here since it efficiently deletes unwanted tonal noise at fixed frequencies (500 Hz and 1000 Hz), with an insignificant effect on neighboring audio, and is computationally lightweight.

D.

Compare the filtered audio to the original and noisy versions. How effective was your filter in improving sound quality?

After applying the notch filters to remove 500 Hz and 1000 Hz tones, the filtered audio is much closer to the original. The annoying hums are removed or significantly reduced.

The rest of the audio remains mostly unchanged because notch filters target only the narrow frequency bands where noise exists.

Overall sound quality is improved, with fewer distractions and clearer content.

Some minor artifacts might occur if the filter bandwidth is too wide, but with proper design, these are minimal.

E.

How did the frequency spectrum change after applying the filter?

The peaks at 500 Hz and 1000 Hz are significantly reduced or gone.
The rest of the spectrum stays mostly the same.
This indicates that the notch filter successfully removed the unwanted noise frequencies while preserving the rest.